Hodge on $X \times X$ Implies Grothendieck's Lefschetz Standard Conjecture B: Lean 4 Formalization

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Abstract

We record a clean, zero-sorry Lean 4 module (StandardConjectures.lean) formalizing the classical implication between the rational Hodge conjecture on the self-product $X \times X$ of a smooth projective variety and Grothendieck's Lefschetz Standard Conjecture $B(X)$. Specifically, let Λ denote the inverse Lefschetz operator L^{-1} on primitive cohomology viewed as a correspondence class on $X \times X$. If Λ is a rational Hodge class and the Hodge conjecture holds on $X \times X$ in the degree of Λ, then Λ is algebraic (LefschetzB Λ). We explicitly separate this implication from the trivial algebraicity of the forward Lefschetz hyperplane cup operator $L = \smile h$ (L_is_algebraic) and document that neither Grothendieck's Conjecture B nor the Clay Hodge Conjecture is unconditionally discharged here.

1 Formalized Theorem Statement

Theorem 1 (hodge_on_product_implies_lefschetz_B)**.** *Let* Λ : Class *represent the Künneth/correspondence class of the inverse Lefschetz operator on* $X \times X$*. Under the hypotheses* hΛ : IsHodge(Λ) *and* hHC : $\forall \gamma$, IsHodge(γ) $\to$ IsAlgebraic(γ) *on* $X \times X$*:*

$$\text{IsHodge}(\Lambda) \ \wedge \ \text{HodgeOnProduct} \implies \text{LefschetzB}(\Lambda).$$

2 Scope and Provenance

- **Origin Commit Timestamp:** Tuesday, September 22, 2026, 18:44:22 CDT (2026-09-22T23:44:22Z).
- **Scope Separation:** $L = \smile h$ is already algebraic (hyperplane section); that is not Standard Conjecture B. Standard Conjectures A, C, and D and unconditional Hodge on $X \times X$ are not claimed.

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